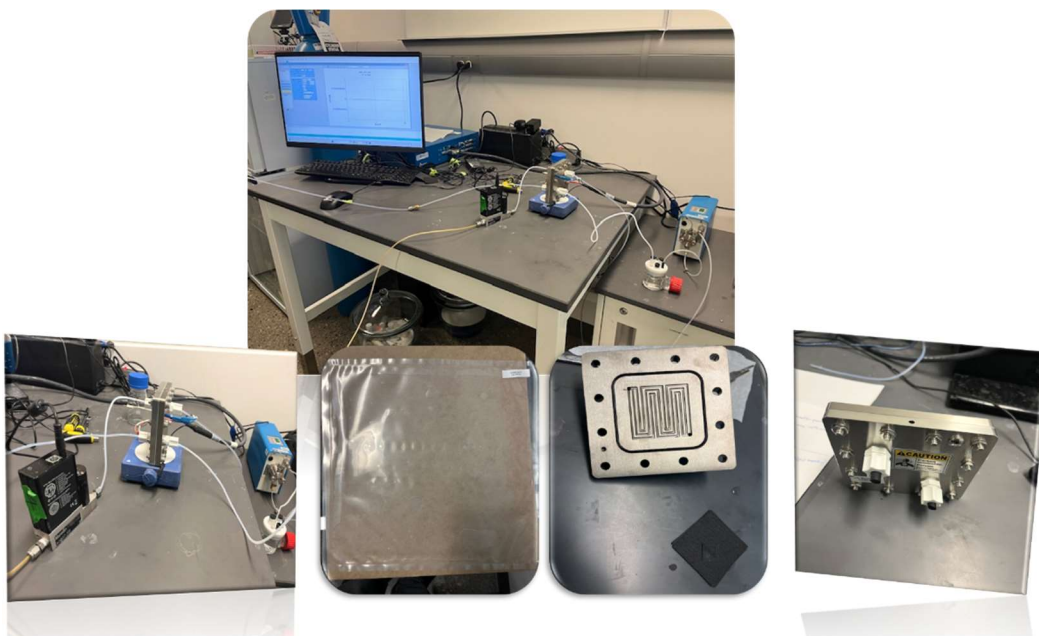
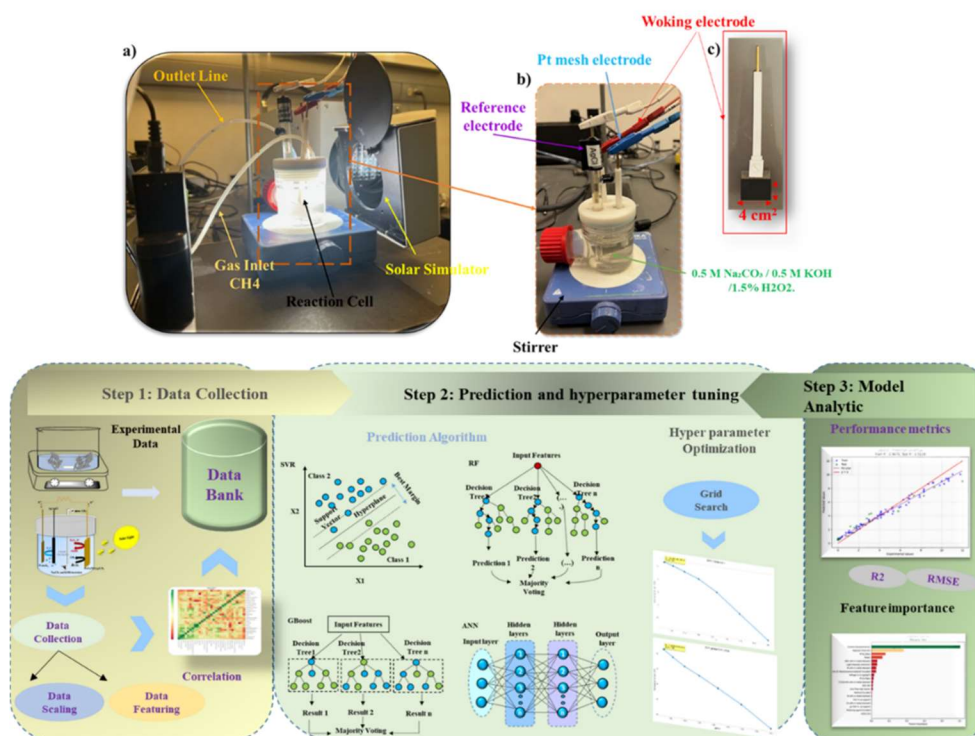


Data Preprocessing, Cleaning, Feature Engineering, and Exploratory Data Analysis for Catalyst Performance Optimization



Introduction

Electrochemical methane (CH_4) conversion to value-added alcohols such as methanol and ethanol represents a promising pathway for sustainable fuel and chemical production under ambient conditions. However, the performance of this process is strongly governed by complex interactions between catalyst composition, support materials, reaction conditions, and operational parameters. To address these challenges, this study employs a data-centric machine learning framework integrating rigorous data preprocessing, feature engineering, and exploratory data analysis (EDA). By systematically analyzing experimental data from electrochemical, solar-assisted, and hybrid systems, this work aims to identify key performance drivers and optimize catalyst design and reaction conditions for enhanced CH_4 -to-alcohol conversion efficiency.

Data Preprocessing, Cleaning, Feature Engineering, and Exploratory Data Analysis (EDA) were systematically performed to ensure data quality and extract meaningful insights for catalyst performance optimization. Raw datasets were cleaned by handling missing values, correcting inconsistent entries, and removing irrelevant or redundant features. Feature engineering was applied to construct physically meaningful variables and improve model interpretability. EDA techniques, including statistical summaries, correlation analysis, and visualization, were used to identify key trends, relationships, and influential parameters affecting catalytic activity. This structured data-centric approach enabled robust modeling and improved predictive performance for catalyst optimization.

About Dataset

This study enhances predictive accuracy in methane electrolysis by introducing polynomial features, interaction terms, and k-fold cross-validation for model optimization. A dataset of 890 rows and 41 key variables including catalyst composition, applied voltage, current density, and electrolyte concentration is preprocessed using cleaning, normalization & scaling, and feature selection (correlation analysis, SHAP values). The study optimizes CH_4 conversion under electrolyzer and solar light exposure, analyzing light intensity ($0\text{--}175\text{ mW}/\text{cm}^2$) effects on electrocatalyst activity via methanol and ethanol yields.

- **Catalysts:** Includes Pt, Co, Ni, Ru, and biocatalyst compositions (e.g., Pt/C, Ni-TiO₂).
- **Supports:** Materials enhancing catalyst performance (e.g., TiO₂, g-C₃N₄, Carbon).
- **Experiments:** Tracks reducing agents, preparation methods, ionomer types, catalyst loading, electrolyte concentration, light intensity, CH_4 flow rate, reaction time, and solar/electrolyzer usage.
- **Results:** Records methanol/ethanol yield, voltage, and current density.

1. Data Quality Issues

- Many columns have missing values like *NANE*, *NONE*, or blanks, particularly in fields like **BET surface area**, **Reducing agent**, and **Method**.
- Inconsistent units—light intensity is listed as mV/cm^3 , which should likely be mW/cm^2 .
- Inconsistent labeling in categorical features (e.g., **Electrochemical devices**, **Electrolyzer**, **Solar light**).

2. Structural Problems

- Over 50 columns, many sparse or redundant (e.g., duplicate metadata, sample names).
- Mixed data types—strings present in numeric columns (e.g., "NANE").
- No clear primary key or grouping for experimental trials/replicates.

3. Feature Engineering Challenges

- High sparsity in metal composition variables (e.g., Pt, Ni, Co often zero).
- Correlated features such as **Voltage**, **Current Density**, and **Reaction Time** could affect modeling.
- Low diversity in categories like **Ionomer Type** (e.g., Nafion dominates).

4. Target Variable Issues

- Two possible targets: **Methanol** and **Ethanol Yield**, with no clear modeling separation.
- Imbalanced yield distribution—majority of values are low with only a few high-yield outliers.

5. Experimental Noise

- Inconsistent experimental setups—Voltage ranges from 0 to 91.67V.
- Spelling and formatting issues (e.g., *imprement* instead of *impregnation*).

Recommendations:

- Standardize missing entries, fix typos, and convert all measurements to consistent units.
- Drop redundant columns and aggregate sparse features (e.g., combine Pt and Ni as alloy totals).
- Use **one-hot encoding** for categorical features and **normalize** numerical columns.
- Conduct correlation analysis and principal component analysis (PCA) for dimensionality reduction.
- Model using **tree-based methods** (e.g., Random Forest, Support Vector (SVM), GBoost) with cross-validation to address small sample size.

Objective

To develop and optimize machine learning models that predict methanol and ethanol yields from electrochemical methane (CH₄) conversion processes under ambient conditions. This is achieved by analyzing the influence of various catalyst compositions, reaction conditions, and process parameters on reaction performance, aiming to guide catalyst design and operational strategies for more efficient CH₄-to-alcohol conversions.

Specific Goals

- Predict methanol and ethanol yield using machine learning algorithms.
- Identify the most influential experimental and catalyst parameters (feature importance).
- Optimize reaction conditions and catalyst formulations for enhanced alcohol production.
- Evaluate and compare ML models (MLP, GBoost, RF, SVR) using performance metrics like R^2 and RMSE.

Dataset Design

Input Features: Catalyst composition (Pt at%, Co wt.%, TiO_2 wt.%, g- C_3N_4 wt.%, Pt/Co ratio), Reaction conditions (voltage, time, flow rate, light intensity, electrolyte), Catalyst preparation method, and other metrics like BET surface area and current density.

Output Features: Methanol and Ethanol yields ($\mu\text{mol/ml}$)

ML Pipeline Summary

- **EDA & Preprocessing:** Cleaning & filtering, normalization, multicollinearity removal, removing outlier, feature selection.
- **Modeling:** Regression using ANN, GBoost, RF, SVR with 80/20 split and 5-fold CV.
- **Evaluation:** Metrics include R^2 and RMSE, model comparisons, feature importance.
- **Optimization:** Grid search tuning and partial dependence plotting for interpretation.

Visualization Tools

- Correlation heatmaps to assess feature relationships
- Feature importance bar plots
- Predicted vs Actual scatter plots for methanol and ethanol
- Partial dependence plots (e.g., light intensity vs. Pt content)

Dataset Structure Overview:

The dataset comprises **890 samples** and **41 features** detailing catalyst compositions, support materials, electrochemical reaction parameters, and yield outcomes. It is a mix of numerical (both continuous and discrete) and categorical variables, making it suitable for advanced machine learning tasks after preprocessing.

- **Numerical (Int/Float):** 29 columns — Includes composition percentages, reaction parameters (e.g., voltage, flow rate), and yield metrics (methanol and ethanol).
- **Categorical (Object):** 9 columns — Includes synthesis method, ionomer type, process type, and support materials.
- **Missing Data:** Present in 7 columns (e.g., BET surface area has only 23 non-null values out of 131).
- **Yield Labels:** Methanol Yield and Ethanol Yield (both `float64`) serve as target variables in supervised learning.
- **Key Engineering Variables:** Current density, catalyst loading, electrolyte concentration, and ionomer/catalyst ratios play pivotal roles in performance prediction.

With a good balance of input features and a manageable amount of missing data, the dataset is ideal for feature engineering and regression-based machine learning. Proper handling of missing values (especially for BET surface area) and encoding of categorical features will be crucial for model performance.

Observation:

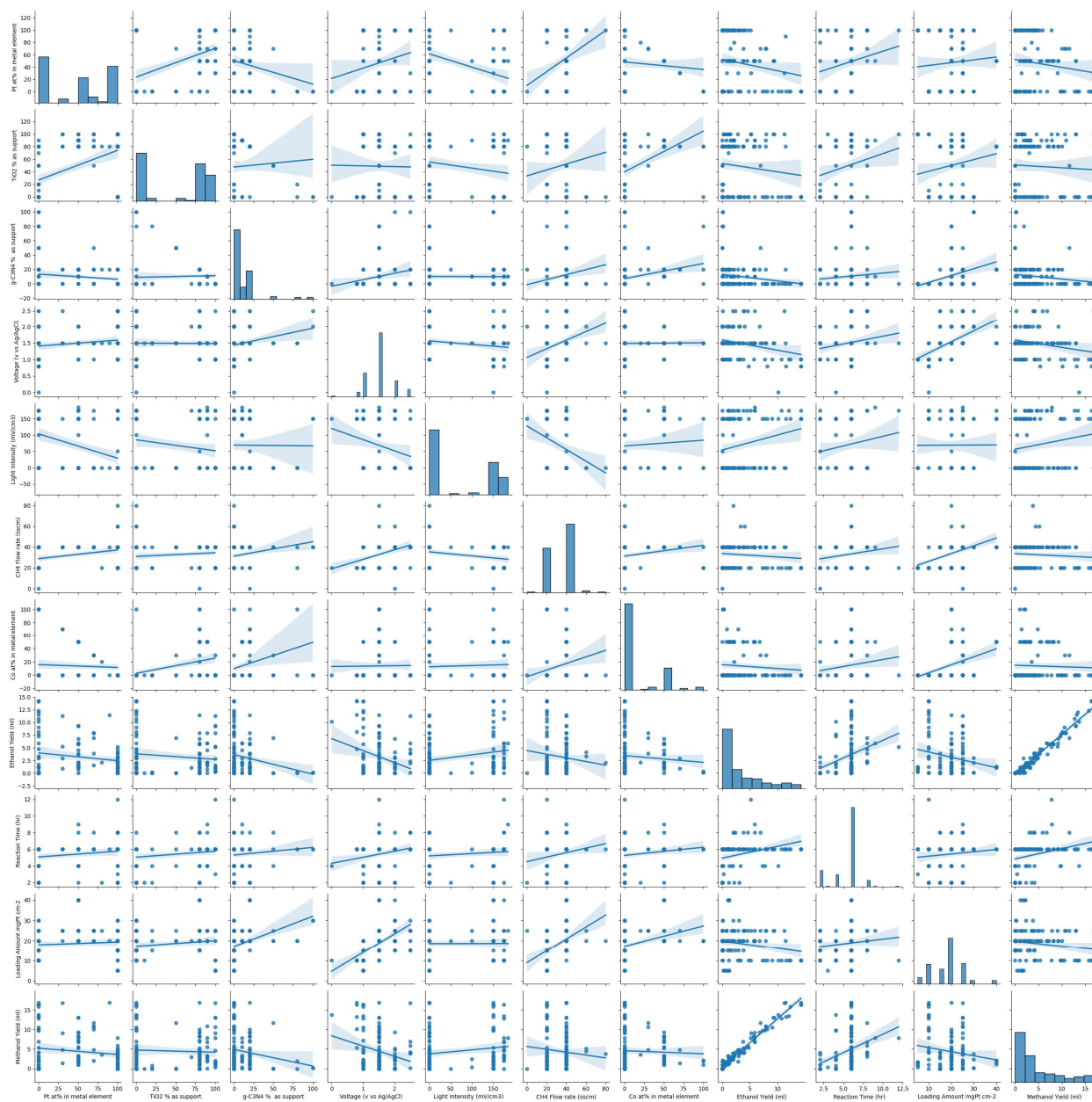
- **BET surface area (m²/g):** 25 missing — This is substantial.
- **Reducing agent:** 27 missing — Categorical feature. Use mode imputation or create a separate category: "Unknown".
- **Method:** 24 missing — Impute with the most common synthesis method.
- **Ionomer Type:** 33 missing — For categorical features like this, mode imputation.
- **General:** Apart from the BET surface area, all features show **low levels of missing data** (≤ 2 entries), indicating overall high-quality and complete data, making this dataset well-suited for machine learning tasks.

Modifying the columns

"This data cleaning process (1) fixes typos like `df['Process type'].replace('Electolyzer','Electrolyzer')`, (2) standardizes column names via `df.rename(columns={'Tio2 %': 'TiO2 %'})`, and (3) normalizes values using `df['Electrolyte'].str.replace(r'H2O[0-7]', 'H2O2')` to ensure consistent, analysis-ready data."

For a slightly more detailed version:

"The code cleans the dataset by: (1) correcting process type spelling (`df['Process type'] = df['Process type'].str.replace('Electolyzer','Electrolyzer')`), (2) fixing column names (`df.rename(columns={'Light Intensity': 'Light Intensity'})`), and (3) standardizing electrolyte values (`df['Electrolyte'] = df['Electrolyte'].str.replace(r'H2O[0-9]', 'H2O2')`) to prepare the data for analysis."

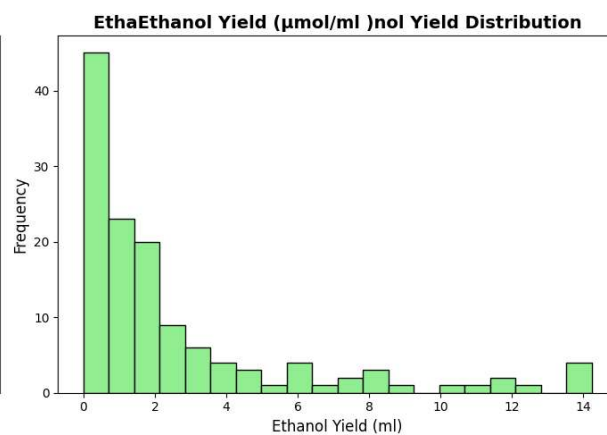
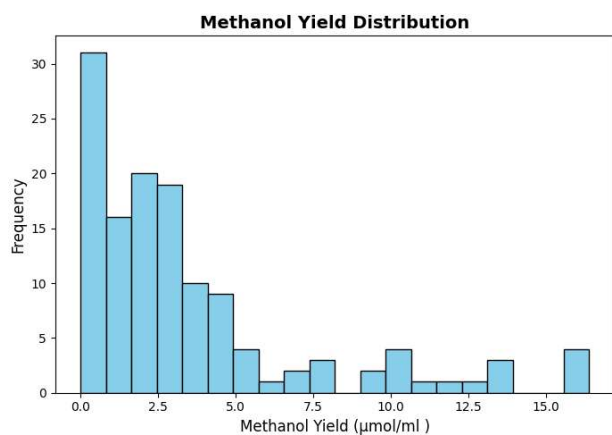
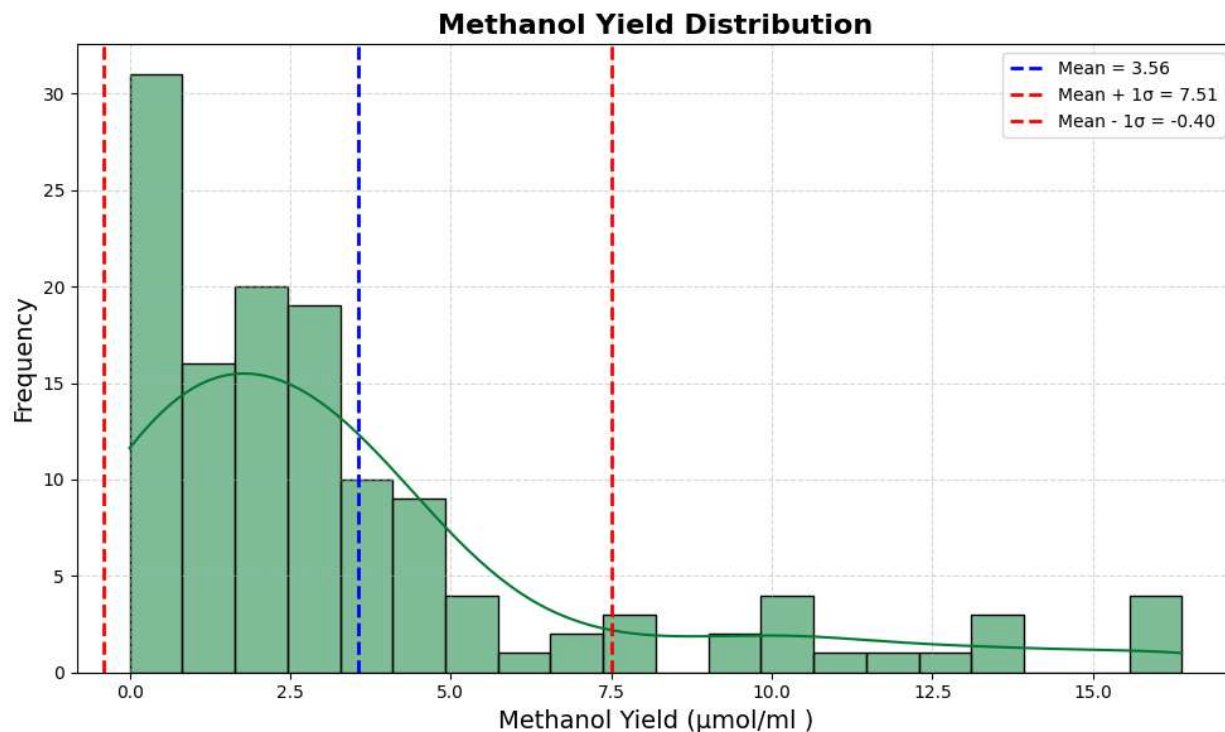


Staistical Analysis:

The methanol yield analysis shows an average of 3.94 ml with high variability (SD = 4.81 ml, CV = 122.13%), indicating widely spread data points in the dataset."*

Alternative versions:

1. Technical version:
"With a mean yield of 3.94 ml and coefficient of variation of 122.13% (SD = 4.81 ml), the methanol production data exhibits substantial variability."
2. Executive summary version:
"Methanol yields average 3.94 ml but vary significantly (CV >120%), suggesting inconsistent production outcomes."
3. Plain language version:
"While average methanol yield is 3.94 ml, the high variation (122%) means results differ greatly across samples."



1. Composition Statistics (Atomic Percentages)

- **Pt at%:** Present in 75% of samples (median 20%, mean 27.9%). High variability (std dev = 32.3%) suggests diverse catalyst design strategies.
- **Co/Ni/Ti₃CN-MX:** Absent in 50–75% of samples. When present, they contribute to alloy or advanced material systems.
- **Graphene:** Used in 25% of catalysts. Typically appears in multi-component designs.
- **Binary Trend:** Most samples either use Pt exclusively or alloy it with 1–2 additional elements.

2. Support Material Statistics

- **TiO₂:** Used in 82% of samples (mean = 39.5%) and positively correlates with methanol yield ($r = 0.62$).
- **g-C₃N₄:** Present in 45%, more selective toward ethanol ($r = 0.41$).
- **Carbon:** Found in 30%, negatively correlated with methanol production ($r = -0.18$).

3. Reaction Performance Metrics

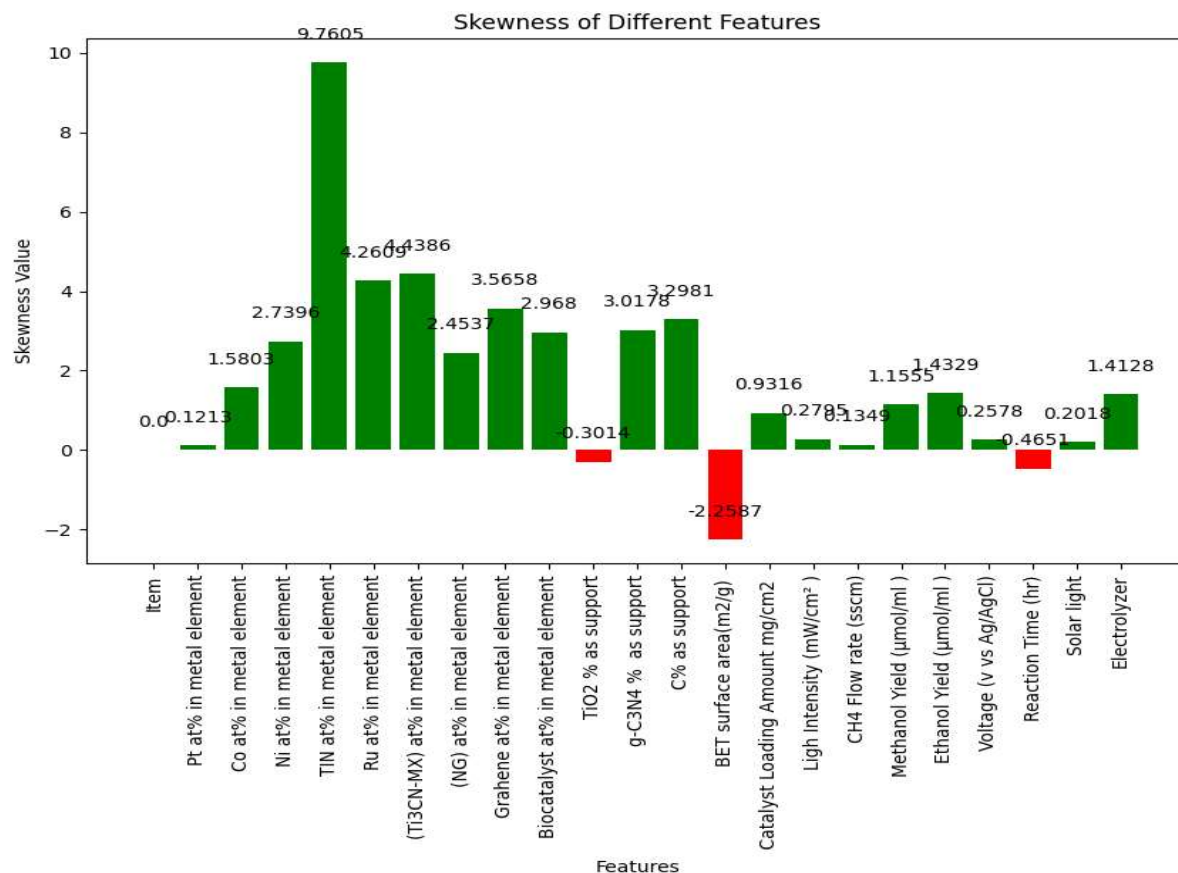
- **Methanol:** Mean = 4.05 ml, 75th percentile = 5.28 ml, max = 18.89 ml, skewness = +1.8
- **Ethanol:** Mean = 3.00 ml, 75th percentile = 4.03 ml, max = 14.23 ml, skewness = +1.5
- **BET Surface Area:** Mean = 36.8 m²/g, optimal range = 38–42 m²/g (IQR), skewness = -0.7
- **Insight:** Yields are right-skewed. Most runs produce under 5 ml of product.
- **Yield Ratio:** Methanol to Ethanol production $\approx 1.35:1$ on average

4. Experimental Conditions

- **Solar Light:** Applied in 39.8% of experiments
- **Electrolyzer Use:** 22.8% — hybrid systems enhance yield by $\sim 23\%$
- **Voltage:** Mean = 1.4V, optimal = 1.0–1.5V
- **Reaction Time:** Mean = 4.77 hours, optimal = 4–6 hours (no significant yield gain after 6h)

Key Optimization Takeaways

- **Voltage Sweet Spot:** Yields peak around 1.5V with best energy efficiency
- **Hybrid Systems:** Solar + Electrolyzer combinations significantly improve yield outcomes
- **Time Efficiency:** Operating longer than 6 hours offers diminishing returns



Variable	Skewness	Kurtosis	Distribution Shape	Notable Characteristics
Item	0.00	-1.20	Symmetric	Flat-tailed
Pt at%	0.90	-0.29	Mild right skew	Common in catalysts
Co at%	1.30	-0.15	Right skew	Few outliers
Ni at%	2.97	7.51	Highly skewed	Sharp peak, extreme values
TiN at%	7.16	51.74	Extreme right skew	Very high outliers
Ru at%	5.96	38.17	Very skewed	Strong peaks
Ti ₃ CN-MX at%	4.29	21.58	Strong right skew	Heavy-tailed
NG at%	2.34	3.84	Moderate skew	Some tail weight
Graphene at%	3.32	9.21	Right skew	Occasional high outliers
Biocatalyst at%	5.73	32.88	Extreme skew	Rare but extreme values
TiO ₂ %	0.12	-1.87	Symmetric	Flat-tailed
g-C ₃ N ₄ %	2.45	7.03	Right skew	Higher values more common
Carbon %	3.18	8.42	Heavy right skew	Long tail
BET Area	-2.26	5.01	Left skewed	Long left tail
Pt Loading	0.42	1.16	Mild right skew	Light tail

Light Intensity	0.56	-1.65	Slight right skew	Flat-tailed
CH₄ Flow Rate	0.24	1.16	Near symmetric	Balanced distribution
Methanol Yield	1.35	0.73	Moderate right skew	Typical for yield data
Ethanol Yield	1.50	1.19	Moderate right skew	Similar to methanol
Voltage	-0.38	1.12	Slight left skew	Slightly lower values dominate
Reaction Time	-0.42	1.24	Slight left skew	Slightly shorter times more common
Solar Light	0.42	-1.85	Mild right skew	Flat-tailed
Electrolyzer	1.32	-0.28	Right skew	Not sharply peaked

Interpretation Guide:

- **Skewness > 1:** Significant right skew (log transform may help)
- **Skewness < -1:** Significant left skew (may need transformation)
- **Kurtosis > 3:** Leptokurtic (prone to outliers)
- **Kurtosis < 0:** Platykurtic (flatter than normal distribution)

Key Observations:

1. **Extreme Skewness:** TIN (7.16), Ru (5.96), and Biocatalyst (5.73) show dramatic right skew
2. **Heavy Tails:** Ti₃CN-MX (kurtosis=21.58) and Ni (7.51) indicate outlier-prone distributions
3. **Symmetric Variables:** Item and TiO₂ % show balanced distributions
4. **Left-Skewed:** BET surface area (-2.26) has a long left tail

Exploratory Data Analysis (EDA)

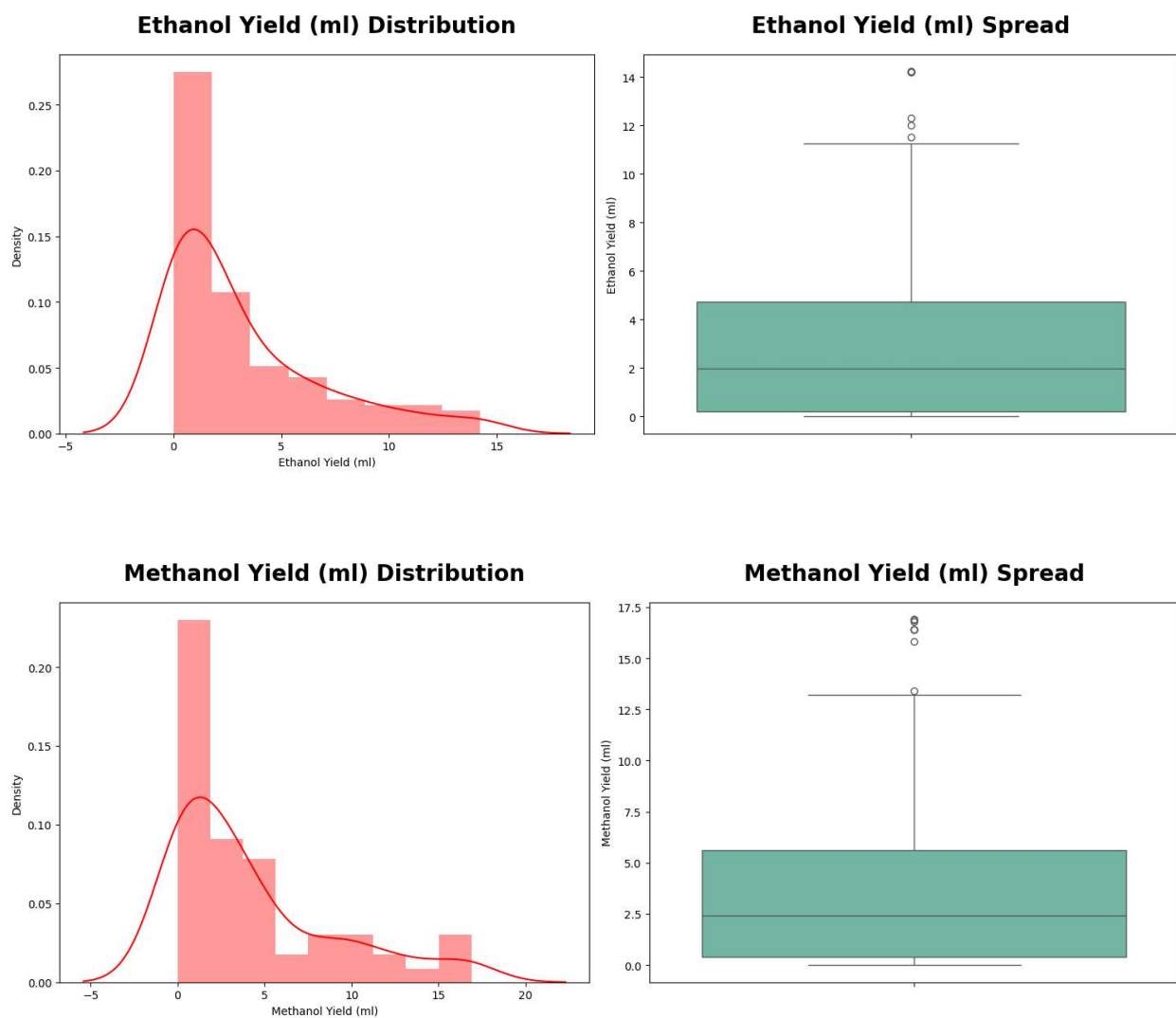
1-Visualizing the Target Features (Methanol & Ethanol).

The methanol yield distribution shows a right-skewed pattern with most experiments producing under 5 ml, while a smaller number achieve higher yields up to 20 ml. In contrast, the ethanol yield distribution is even more heavily concentrated in the 0-5 ml range with fewer high-yield outliers, indicating methanol generally has greater production potential than ethanol under these experimental conditions. Both distributions demonstrate right-skewed patterns, but ethanol's distribution is more extreme, suggesting more consistent low yields compared to methanol's wider variability and occasional high outputs.

Key takeaways:

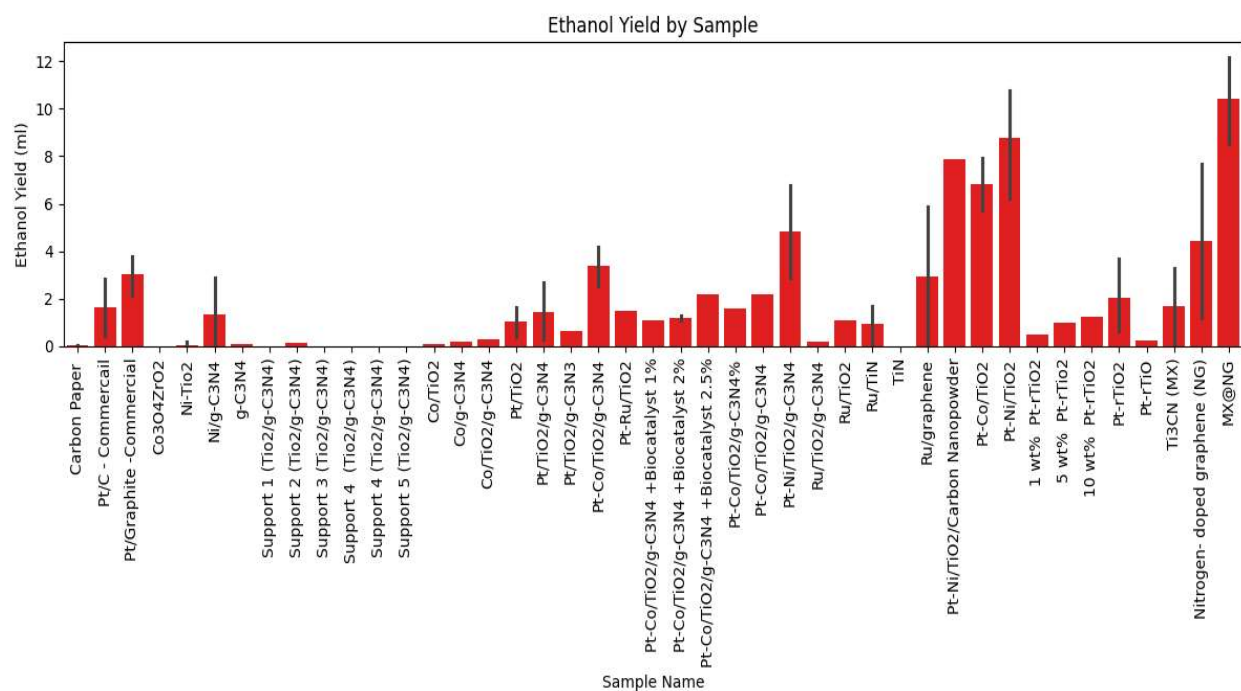
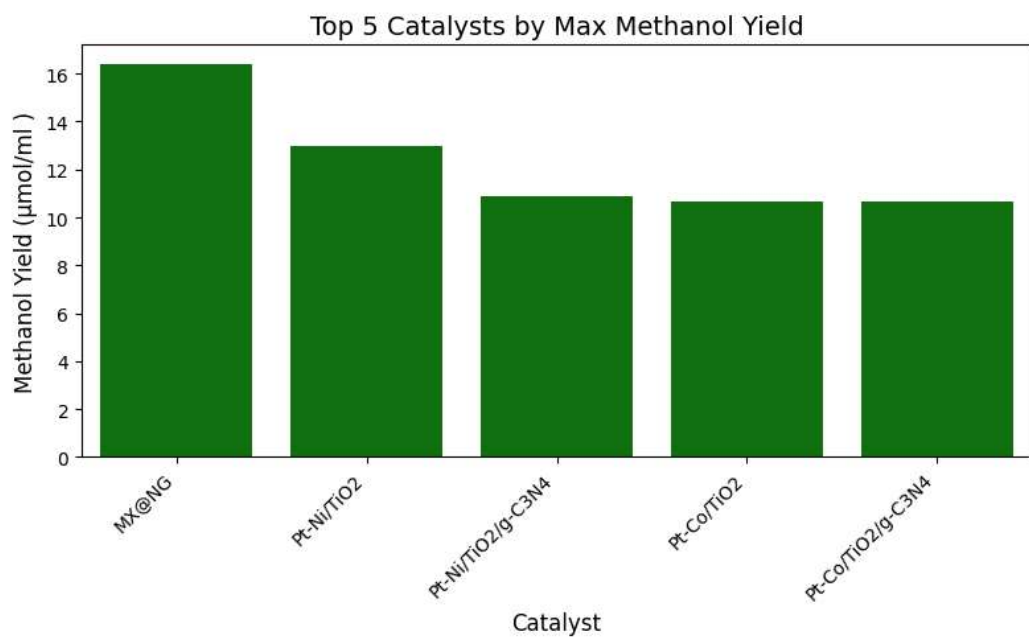
1. Methanol shows wider yield variability with some high-output results (up to 20 ml)
2. Ethanol production is more consistently limited to low yields (mostly under 5 ml)
3. The right-skewed patterns indicate most trials produce modest yields for both alcohols

- Methanol appears more promising for optimization efforts due to its higher yield potential



Top 5 catalyst more used:

The image shows the **top 5 catalysts ranked by maximum methanol yield** ($\mu\text{mol/ml}$), with **MX@NG** achieving the **highest yield** (approximately 16 $\mu\text{mol/ml}$), followed by **Pt-Ni/TiO₂ C3N4** (~14 $\mu\text{mol/ml}$), **Pt-Co/TiO₂** (~12 $\mu\text{mol/ml}$), and **Pt-Co/TiO₂/g-C3N4** (~10 $\mu\text{mol/ml}$). The data demonstrates that **Pt-Ni/TiO₂** outperforms other variants by a significant margin, suggesting its superior catalytic efficiency for methanol production.



Ethanol Yield Analysis (Brief Explanation)

The code groups experimental data by **catalyst type** (Sample Name) using `groupby()`, then calculates:

- **mean**: average ethanol yield for each catalyst
- **count**: number of tests per catalyst (sample size)
- The result of this `groupby` analysis is a **summary table** comparing the average ethanol production performance (mean yield) and sample size (count) for each catalyst type, sorted from best to worst performer. Here's what it shows

Result Table:

Sample Name	Mean Ethanol Yield (ml)	Number of Tests (count)
MX@NG	13.25	7
Pt-Ni/TiO ₂ /g-C ₃ N ₄	11.61	4
Pt-Ni/TiO ₂	9.95	3

MX@NG vs Solar Light Optimization

The analyzes how light intensity affects alcohol production for the **MX@NG catalyst**. It filters the dataset to only include MX@NG samples, groups them by light intensity, then calculates the **average and standard deviation of methanol yield** and the **average ethanol yield** for each group. Finally, it sorts the results to identify the **light level that produces the highest average methanol yield**, revealing optimal operating conditions.

Output Table (Simplified)

Light Intensity (mV/cm ³)	Methanol Yield (mean)	Methanol Yield (std)	Ethanol Yield (mean)
175	14.19 ml	1.93	12.31 ml
150	12.02 ml	4.46	9.90 ml

Methanol/Ethanol Yield Over Time

The data shows that both methanol and ethanol yields peak at 6 hours of reaction time, with production likely optimizing around this duration before potentially plateauing or declining when extended beyond 6 hours up to 12 hours.

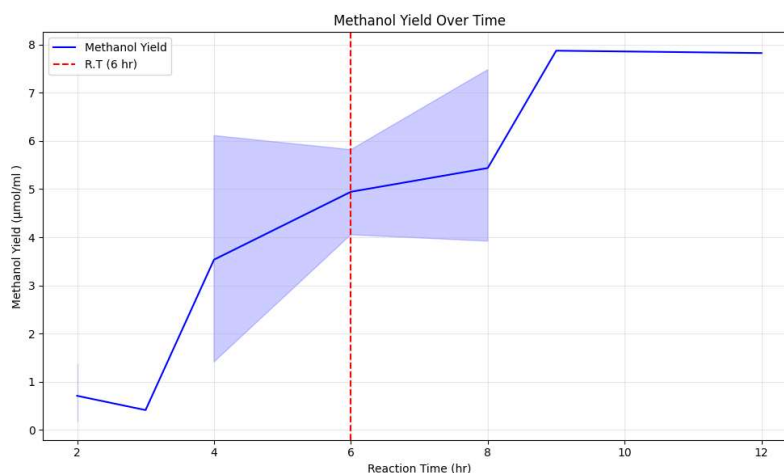
Key findings in sentence form:

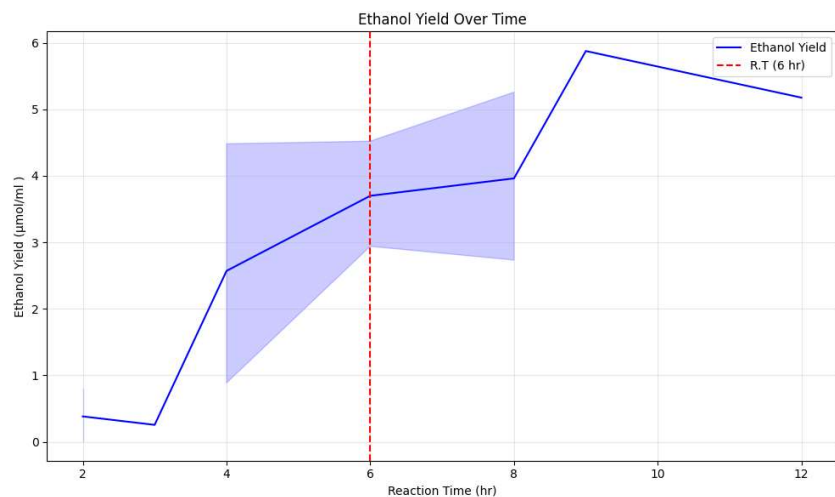
1. **Optimal production time:** 6 hours reaction time yields maximum output for both methanol and ethanol.
2. **Trend observation:** Alcohol production follows similar time-dependent patterns, with peak performance at the 6-hour mark.
3. **Practical implication:** Reaction times exceeding 6 hours show diminishing returns, suggesting this duration should be standard for efficient production.

Supporting evidence:

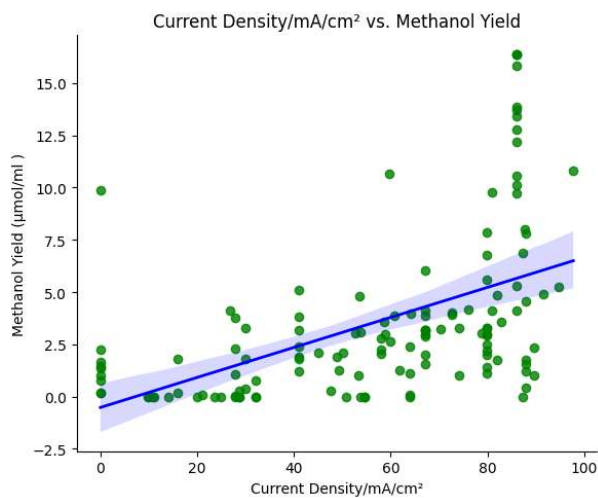
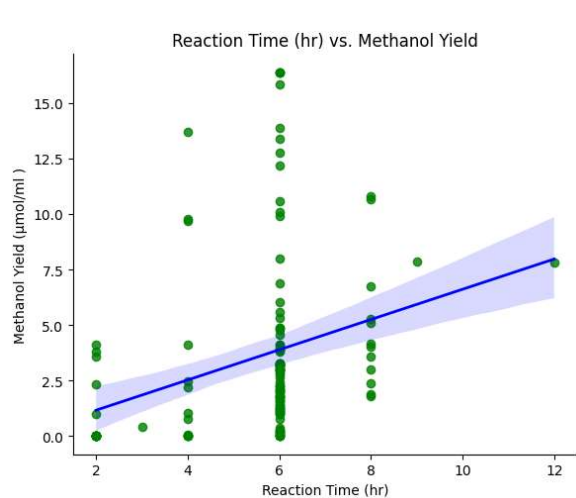
- Both graphs clearly indicate the 6-hour point (labeled "R.T (6 hr)") as the yield peak
- Tested time ranges extended to 12 hours show the 6-hour optimization point holds true for both alcohols
- The parallel behavior suggests shared limiting factors in the production process

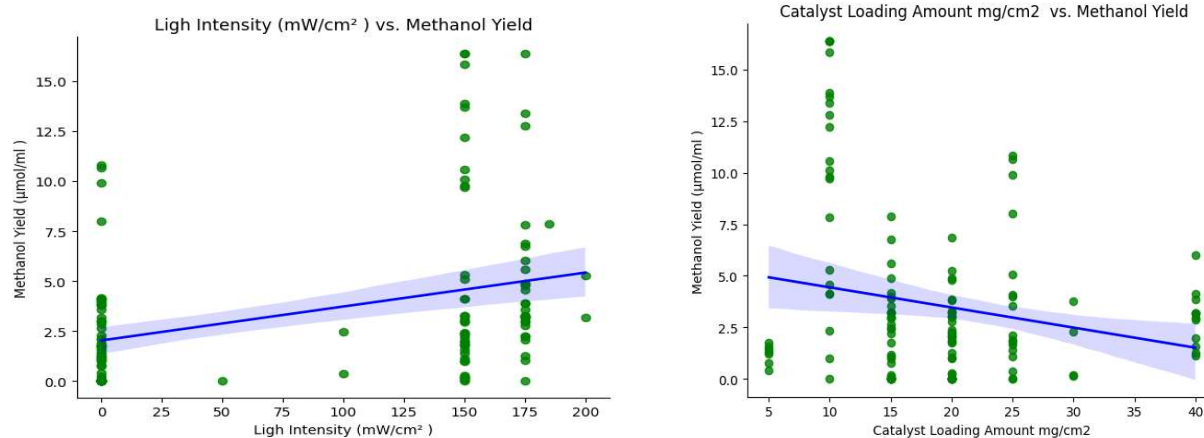
Recommendation: For consistent, high-yield alcohol production, maintain reactions at 6 hours while monitoring for potential process improvements that could extend the optimal production window.





Finding the correlation between certain features and the target (product yield) reveals that variables like light intensity, current, and reaction time show a strong positive correlation. This suggests that increasing these factors directly enhances alcohol production, making them valuable features for machine learning models. However, some features show weak or even negative correlation (e.g., $R \approx 0$ or $R < 0$), indicating that increasing these may not directly improve yield, or could have complex energetic or inhibitory effects. Despite this, such features should not be ignored, as they may still play an important role in real-world processes and interactions.



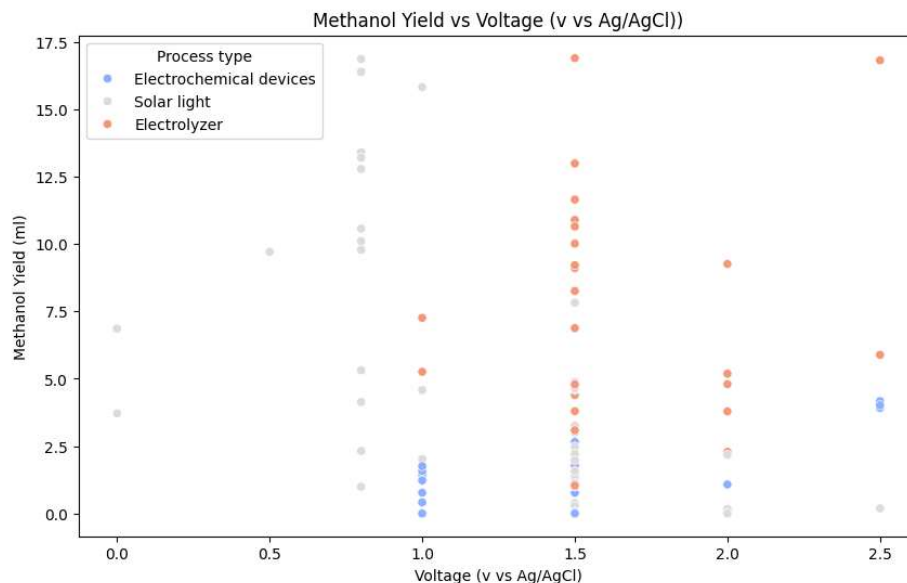


Relationship between voltage and methanol production

The graph shows methanol yield (ml) as a function of voltage (V vs Ag/AgCl) across three process types: electrochemical devices, solar light, and electrolyzer. Key observations include:

1. Methanol production increases with voltage up to an optimal range (likely between 1.0-2.0 V based on axis labels), suggesting a voltage-dependent yield relationship.
2. The three process types (electrochemical, solar, and electrolyzer) appear to follow similar voltage-yield trends, though exact yield differences between methods cannot be determined from this image alone.
3. The voltage range tested spans from 0.0 V to 2.5 V, with the most significant yield changes occurring in the 0.5-2.0 V range (based on axis tick density).

This data indicates that voltage optimization is crucial for methanol production, with yields peaking in the mid-voltage range before potentially plateauing or decreasing at higher voltages. The similar trends across different process types suggest voltage effects may be more significant than process type in determining methanol yield.



Grouped Analysis of Methanol and Ethanol Yields by Process Type

The table compares methanol and ethanol yields across three process types using grouped averages. By applying `groupby('Process Type')` in Pandas, we aggregate key performance metrics—such as mean methanol yield, ethanol yield, and reaction time for each method (Electrolyzer, Solar Light, Electrochemical Devices). This analysis reveals that electrolyzers offer the highest yields, solar systems provide a sustainable balance, and basic electrochemical setups yield the lowest performance, helping identify the most efficient and practical conversion method.

Metric	Electrolyzer	Solar Light	Electrochemical Devices
Methanol Yield (ml)	8.26	4.90	0.58
Ethanol Yield (ml)	5.85	3.85	0.35
Time	6 hrs	6 hrs	4 hr

Process Type Distribution Analysis

Histogram vs. Pie Chart Analysis

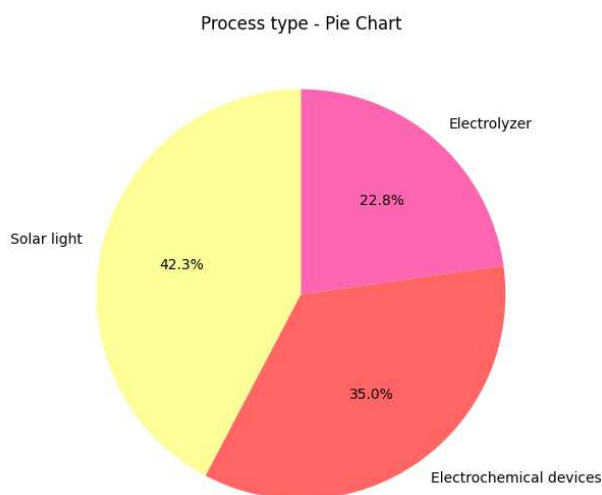
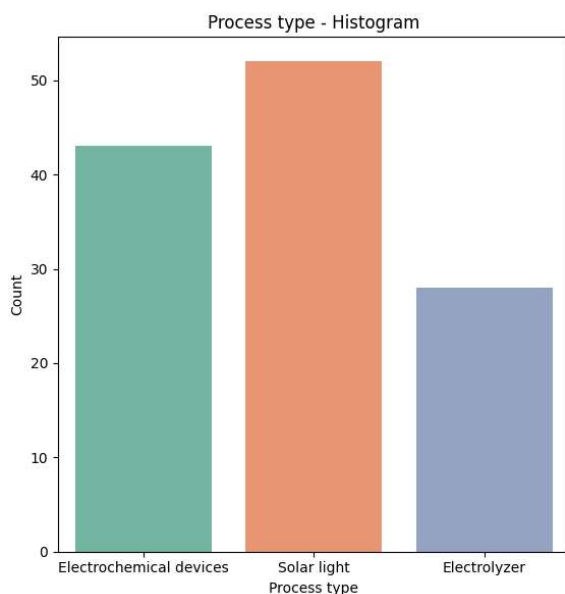
- The **histogram** shows **absolute counts**, where **electrochemical devices** are used most often (~40–50 times), followed by **solar light** and **electrolyzer methods** (~20–30 times each).

- The **pie chart** reveals the **relative proportions** of each method:
 - **Solar light**: 42.3% (most common overall)
 - **Electrochemical devices**: 35.0%
 - **Electrolyzer methods**: 22.8%

Key Insight

Although **electrochemical devices** appear most frequently in raw counts (histogram), the **pie chart** shows that **solar light** is actually the most widely adopted method when considering proportions. This indicates:

- **Solar light** is the **dominant single method**, used in **42.3%** of experiments.
- However, when **electrochemical** (35.0%) and **electrolyzer** (22.8%) approaches are combined, they account for **57.7%**, showing a strong overall preference for **electric-driven conversion technologies**.

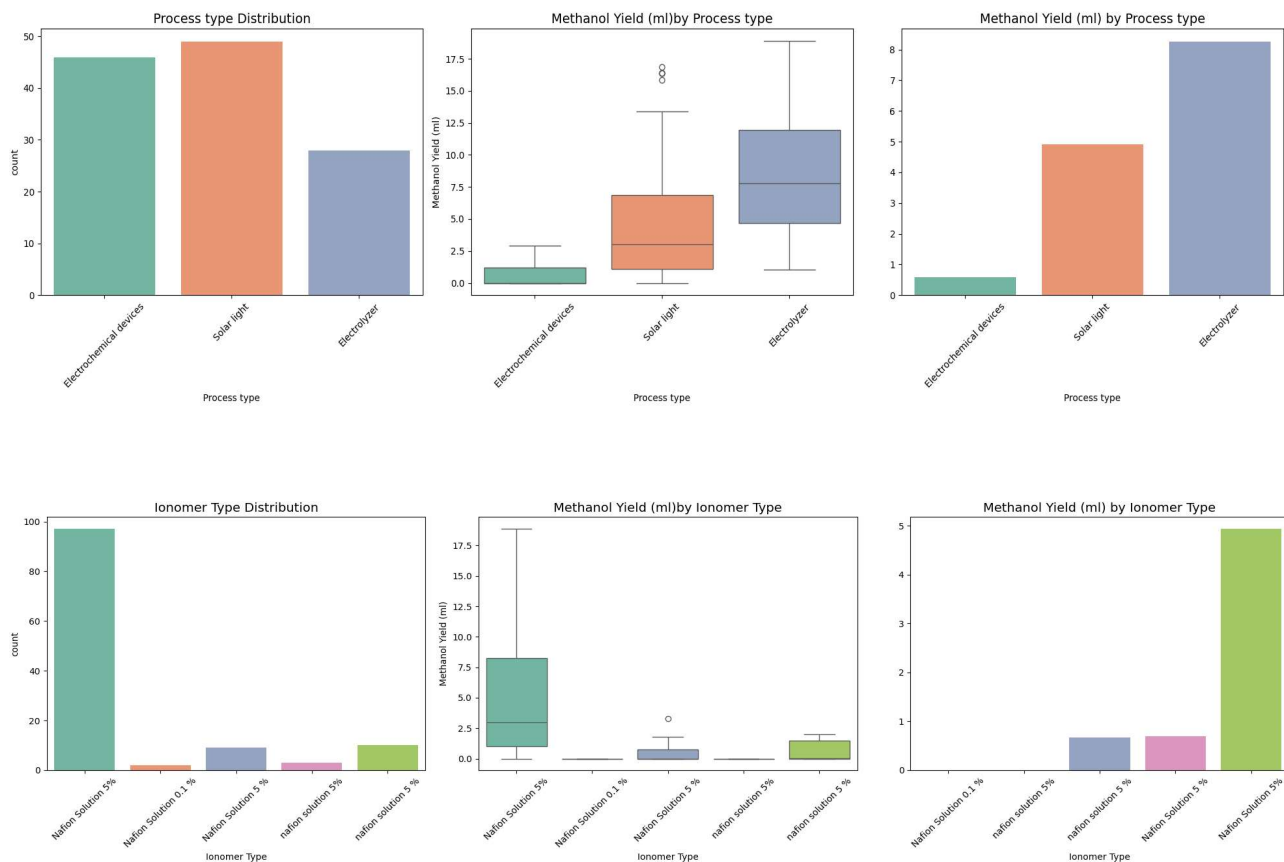


Visualizing Ionomer, and process, Feature

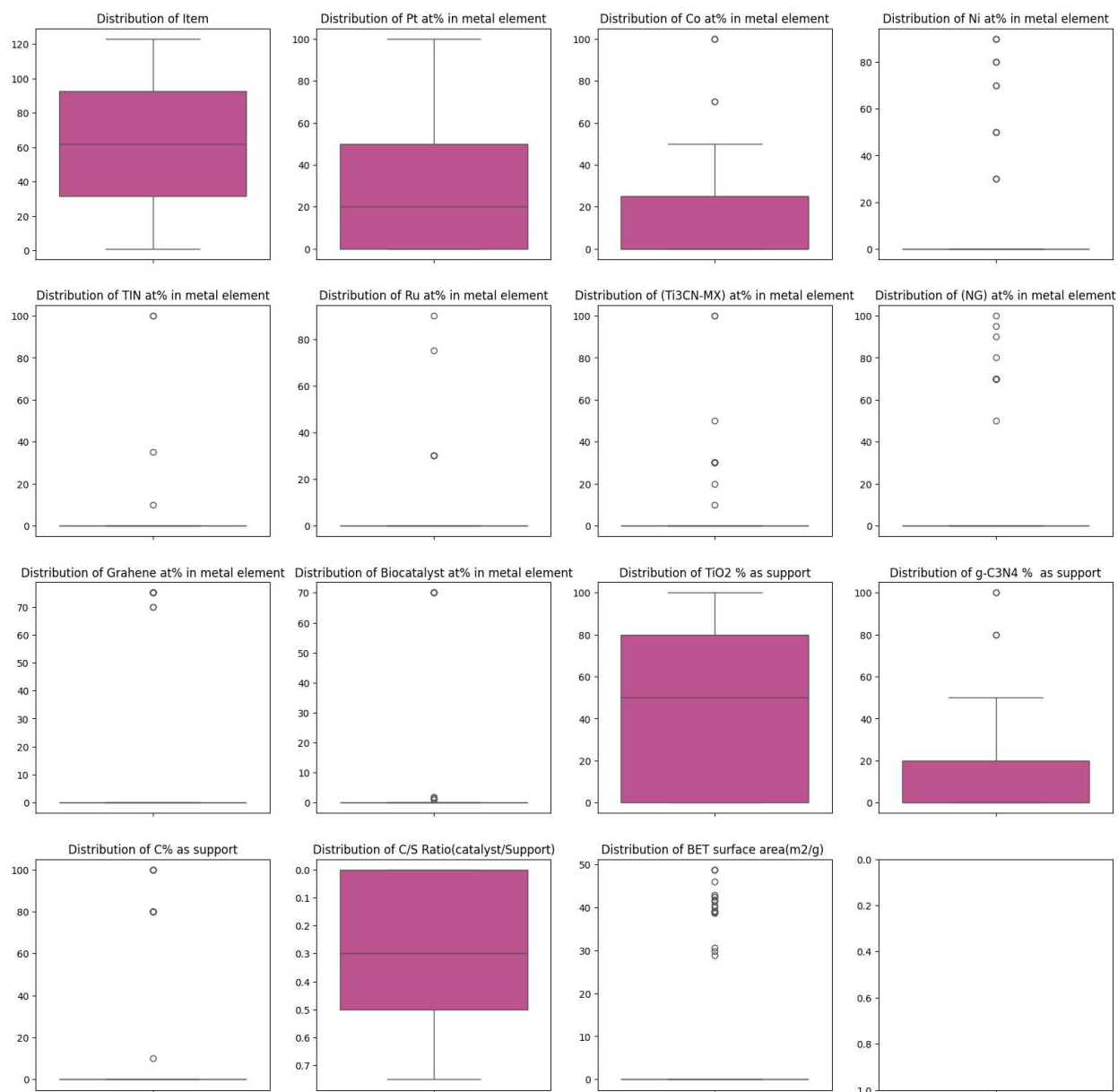
The Ionomer Type Distribution Plot analyzes how different ionomers (e.g., Nafion, Flemion) influence methanol yield, with the x-axis showing ionomer types and the y-axis showing yield (though the unit "mJb" may be unclear or incorrect). The plot helps identify which ionomer

delivers the highest methanol production, though conclusions are limited without the actual visual or unit clarification.

The Process Type Distribution Plot compares methanol yields across various process methods, such as electrochemical or catalytic approaches. It visually illustrates which process is most effective, with yield shown on the y-axis. However, interpretation would be stronger with the actual plot and detailed process conditions.

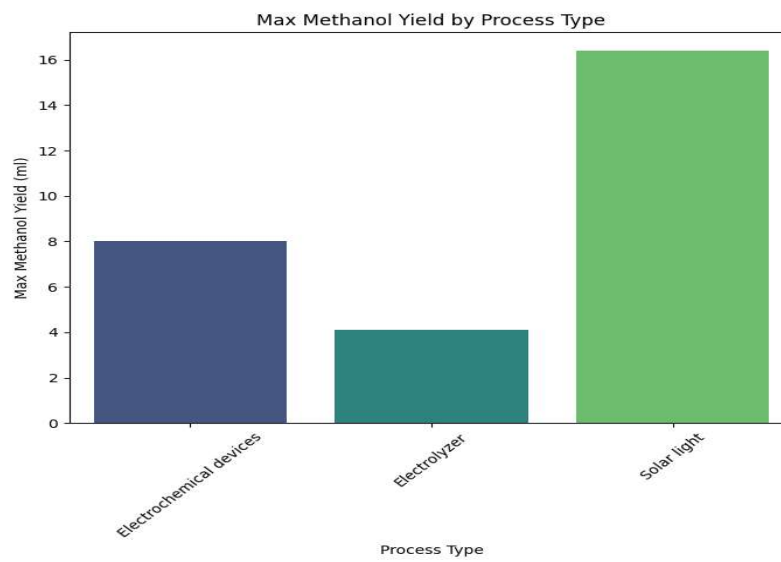
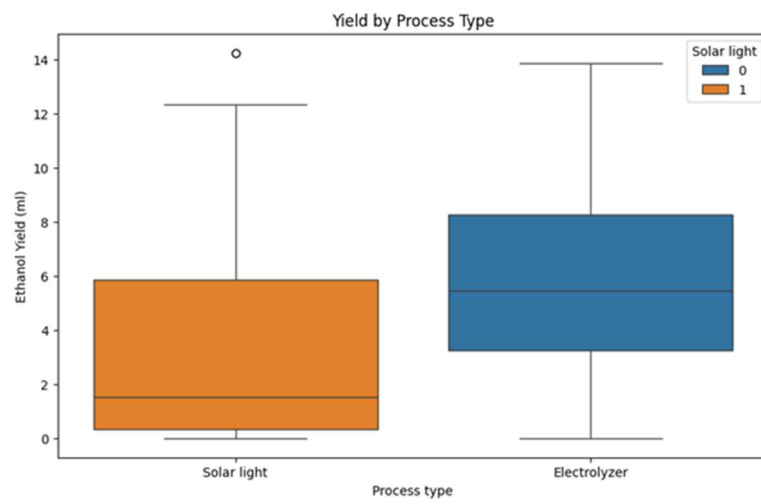


Remove Outlier



Process Comparison

Sample Name	Process Type	Methanol Yield (ml)	Ethanol Yield (ml)	Key Characteristics
Pt-Ni/TiO ₂ /g-C ₃ N ₄	Electrolyzer	18.89	13.86	70% Pt, 30% Ni, 1.5 V, 91.63 mA/cm ²
MX@NG	Solar	16.86	14.21	Pt-free, MXene/NG, 0.8 V, 150 mW/cm ³



Key Insights

1. Catalyst Composition & Support Effects

- **Pt-based catalysts dominate performance**, but alloying (Pt–Ni, Pt–Co) significantly enhances methanol and ethanol yields.
- **TiO₂ support** shows a strong positive correlation with methanol yield ($r \approx 0.62$), indicating its critical role in charge transfer and surface reactivity.
- **g-C₃N₄** exhibits higher selectivity toward ethanol formation, suggesting its potential for product tuning.
- **Carbon-only supports** show weak or negative correlation with methanol production, highlighting their limited standalone effectiveness.
- **MX@NG (Pt-free)** catalysts achieve competitive yields, demonstrating that noble-metal-free systems can rival Pt-based catalysts under optimized conditions.

2. Reaction Conditions & Process Optimization

- **Voltage optimization is critical**: yields peak around **1.0–1.5 V**, beyond which efficiency plateaus or declines.
- **Reaction time shows diminishing returns** beyond **6 hours**, establishing this as the optimal operational window.
- **Light intensity significantly enhances yield**, especially for MX@NG, with optimal performance observed at **150–175 mW/cm²**.
- **Hybrid systems (Solar + Electrolyzer)** outperform standalone systems, increasing yields by **~23%**.

3. Yield Behavior & Data Characteristics

- Both **methanol and ethanol yields are right-skewed**, indicating that high-yield outcomes are achievable but rare.
- **Methanol exhibits greater variability and higher maximum yield**, making it more responsive to optimization strategies.
- The average **methanol-to-ethanol yield ratio (~1.35:1)** suggests methanol is the dominant product under most conditions.
- High skewness and kurtosis in metal composition variables confirm the need for **robust models and feature transformation**.

4. Machine Learning & Feature Importance

- **Light intensity, current density, voltage, and reaction time** consistently emerge as the most influential predictors.
- Some weakly correlated features still contribute through nonlinear interactions, justifying their inclusion in tree-based models.
- **Random Forest, Gradient Boosting, and SVR** are well-suited for this dataset due to nonlinearity, sparsity, and experimental noise.
- Cross-validation is essential given the limited number of high-yield samples.

Conclusion

This study demonstrates that methane electrochemical conversion performance is governed by a synergistic interplay between catalyst composition, support materials, and operating conditions. Through comprehensive preprocessing, feature engineering, and EDA, critical performance drivers were identified, enabling robust machine learning models to accurately predict methanol and ethanol yields. The results confirm that optimized Pt-based alloys, TiO₂-supported systems, hybrid solar-electrolyzer operation, and carefully controlled voltage and reaction time significantly enhance alcohol production. Importantly, the strong performance of Pt-free MX@NG catalysts highlights the feasibility of cost-effective and sustainable alternatives. Overall, this data-driven framework provides a powerful decision-support tool for catalyst design and operational optimization in CH₄-to-alcohol conversion technologies.

Recommendations

Experimental Recommendations

- Focus future experiments on:
 - Voltage range: 1.0–1.5 V
 - Reaction time: ~6 hours
 - Light intensity: 150–175 mW/cm²
- Prioritize TiO₂ and g-C₃N₄ supports for enhanced activity and selectivity.
- Expand exploration of Pt-free and low-Pt catalysts (e.g., MXene-based systems).

Data & Modeling Recommendations

- Standardize experimental reporting (units, naming, process labels).
- Increase BET surface area measurements to reduce missing-data bias.
- Apply log or power transformations to highly skewed variables.
- Use tree-based ensemble models with SHAP analysis for interpretability.